

MY1016Z / KW1016Z FAMILY | 250 W DC GEAR MOTOR

Actual sample: KW1016Z-250W24V | Page 01 rebuilt from supplied source drawing [S] and sample photos [P]

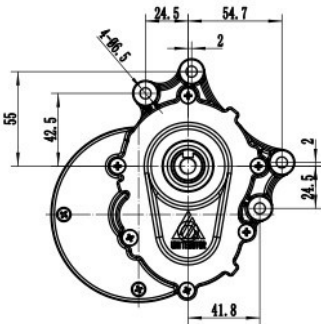
01 GENERAL ARRANGEMENT — RECONSTRUCTED PAGE

SC-MTR-1016Z-REF

REV C / 2026-10-01

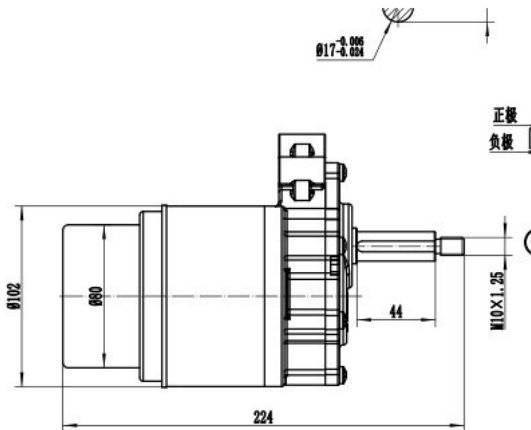
A3 LANDSCAPE / ENGLISH

SUPPLIED SOURCE DRAWING [S] / OUTPUT-END VIEW



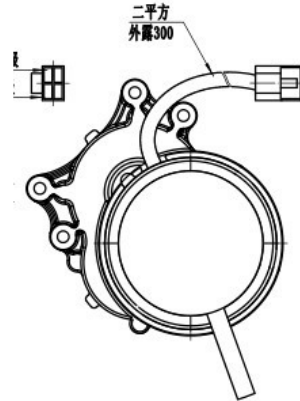
Source variant front view used for shape reference.

SUPPLIED SOURCE DRAWING [S] / SIDE VIEW



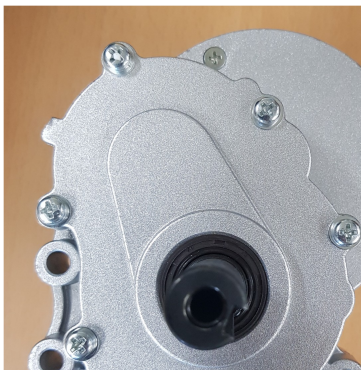
Source variant side view; overall 224 mm not released as actual sample size.

SUPPLIED SOURCE DRAWING [S] / REAR VIEW



Source drawing includes rear-path details not confirmed on the sample.

ACTUAL SAMPLE PHOTO [P] / OUTPUT-END VIEW



Actual sample confirms the output-end housing contour and 4 mounting ears.

ACTUAL SAMPLE PHOTO [P] / SIDE VIEW



Actual sample side view used to correct visual proportion and casing form.

ACTUAL SAMPLE PHOTO [P] / REAR VIEW

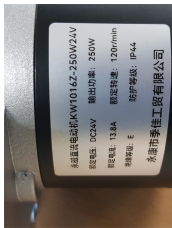


Actual sample rear cover has radial ribs and no visible brake release lever.

ACTUAL-SAMPLE DATA AND SHAPE RECONCILIATION

Model	KW1016Z-250W24V actual sample; MY1016Z family-equivalent outer form [P/S]
Nameplate	24 V DC / 250 W / 13.8 A / 120 r/min / Class E / IP44 [P]
Visual corrections	Page 01 now follows the supplied photos for front, side, and rear external form.
Source-only items	224 mm overall length, M10×1.25 callout, and some rear-path details remain source-only [S/V].
Applicability	Two purchased motors are same-handed. This document keeps one common motor page only.

NAMEPLATE EVIDENCE [P] AND PAGE-01 NOTES



- 1) This rebuilt page prioritizes visual fidelity to the actual sample photos [P].
- 2) The supplied source drawing [S] remains valuable for family geometry and shaft-family reference.
- 3) Rear-view source details absent from the real sample are not treated as released sample features.
- 4) Use Sheets 02 and 03 as reference only until caliper values are entered on Sheet 05.

[P] PHOTO / NAMEPLATE [S] SUPPLIED SOURCE VARIANT [D] DERIVED [V] VERIFY ACTUAL

Units: mm unless noted. This page is a reference/reconciliation sheet, not a machining release by itself.

No caliper measurement record was available when this page was regenerated; external proportions were rebuilt using the supplied photos.

Use the actual sample, not this page, as the authority for final bracket clearance and interference decisions.

GENERAL ARRANGEMENT

Page 01 rebuilt from source drawing and real sample photos

DRAWING	REVISION	FORMAT	SHEET	
SC-MTR-1016Z-REF	C	A3	1 / 5	
VIEW SCALE: NTS	UNITS: mm	LANG: EN	QTY: 2	

MY1016Z-FAMILY / 250 W DC GEAR MOTOR

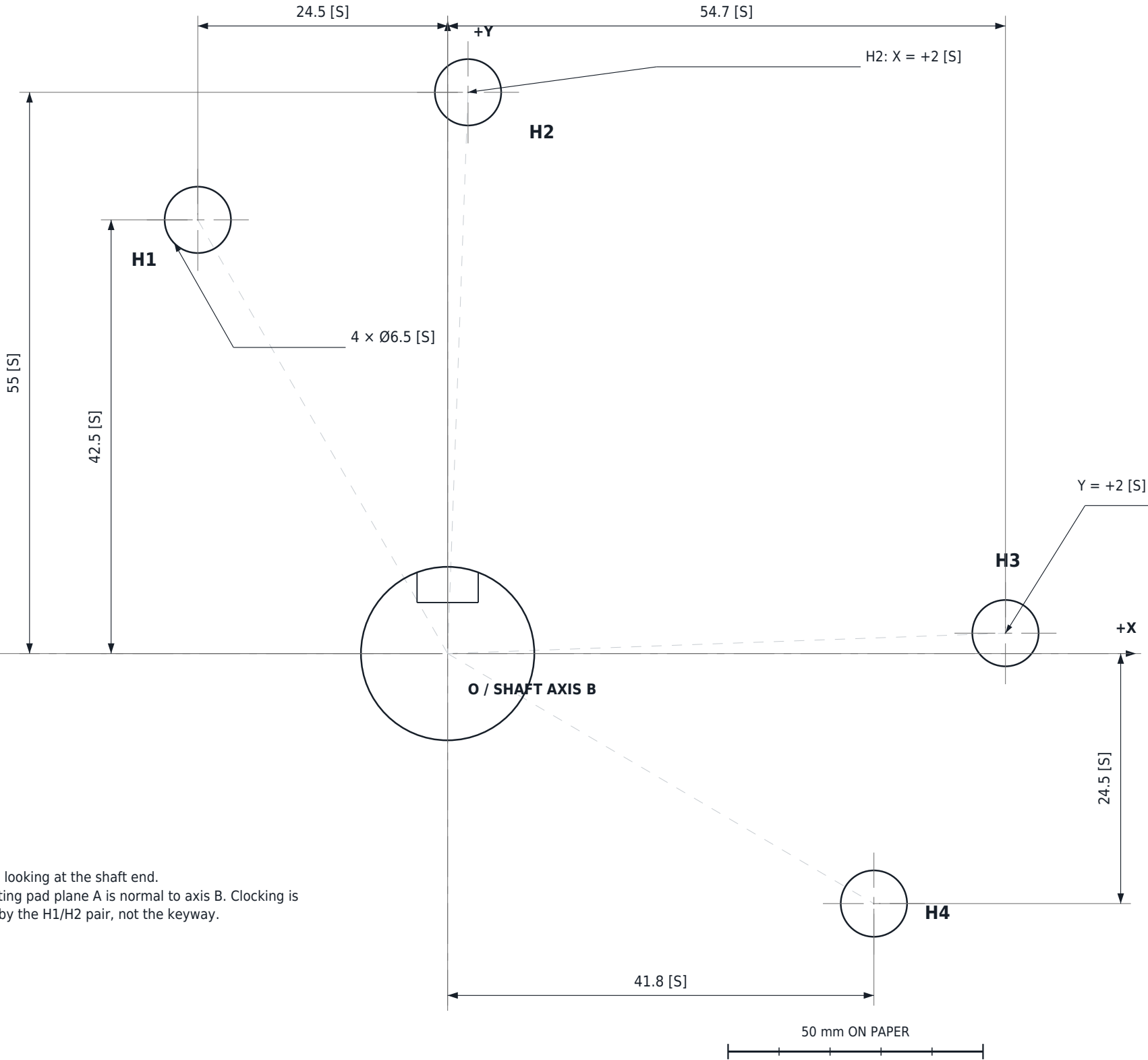
Actual sample: KW1016Z-250W24V | ONE COMMON HAND / QUANTITY 2

02 MOUNTING INTERFACE / SOURCE NOMINAL GEOMETRY

SC-MTR-1016Z-REF

REV A / 2026-10-01

A3 LANDSCAPE / ENGLISH



SOURCE NOMINAL HOLE COORDINATES

HOLE	X	Y	BORE
H1	-24.5	+42.5	Ø6.5 [S]
H2	+2.0	+55.0	Ø6.5 [S]
H3	+54.7	+2.0	Ø6.5 [S]
H4	+41.8	-24.5	Ø6.5 [S]

Coordinates transcribe the supplied drawing [S]; origin and signs are defined above. H4 Y is referenced to the shaft horizontal centerline, not H3.

INDEPENDENT CENTER-DISTANCE CHECKS [D]

PAIR	DISTANCE	STATUS
H1 - H2	29.30	REF only
H2 - H3	74.74	REF only
H3 - H4	29.47	REF only
H4 - H1	94.26	REF only
H1 - H3	88.95	REF only
H2 - H4	88.91	REF only

Calculated from the nominal coordinates: $c = \sqrt{(\Delta X^2 + \Delta Y^2)}$. These are redundant checking dimensions, not new toleranced requirements.

HOLD POINT - BRACKET INTERFACE

Confirm all four bores, six center distances, pad coplanarity and clocking on both actual motors. Cover screws are not mounting bores. Ø6.5 is a bore diameter, not an M6 thread. Do not replace this pattern with the earlier approximate 30 / 73.5 mm values.

[P] PHOTO / NAMEPLATE [S] SUPPLIED SOURCE VARIANT [D] DERIVED [V] VERIFY ACTUAL

Units: mm. No general tolerance is assigned. Do not scale NTS views or photographs.

REFERENCE / VERIFICATION REQUIRED - NOT RELEASED FOR MACHINING

No specimen caliper readings were supplied. [S] values are not measured KW1016Z dimensions.

Both purchased motors use this same part definition; installation rotation is an assembly task.

MOUNTING INTERFACE

Single-hand motor reference / Smart-Cart Capstone

DRAWING SC-MTR-1016Z-REF	REVISION A	FORMAT A3	SHEET 2 / 5
VIEW SCALE: 2:1 (interface)	mm	EN	QTY 2

MY1016Z-FAMILY / 250 W DC GEAR MOTOR

Actual sample: KW1016Z-250W24V | ONE COMMON HAND / QUANTITY 2

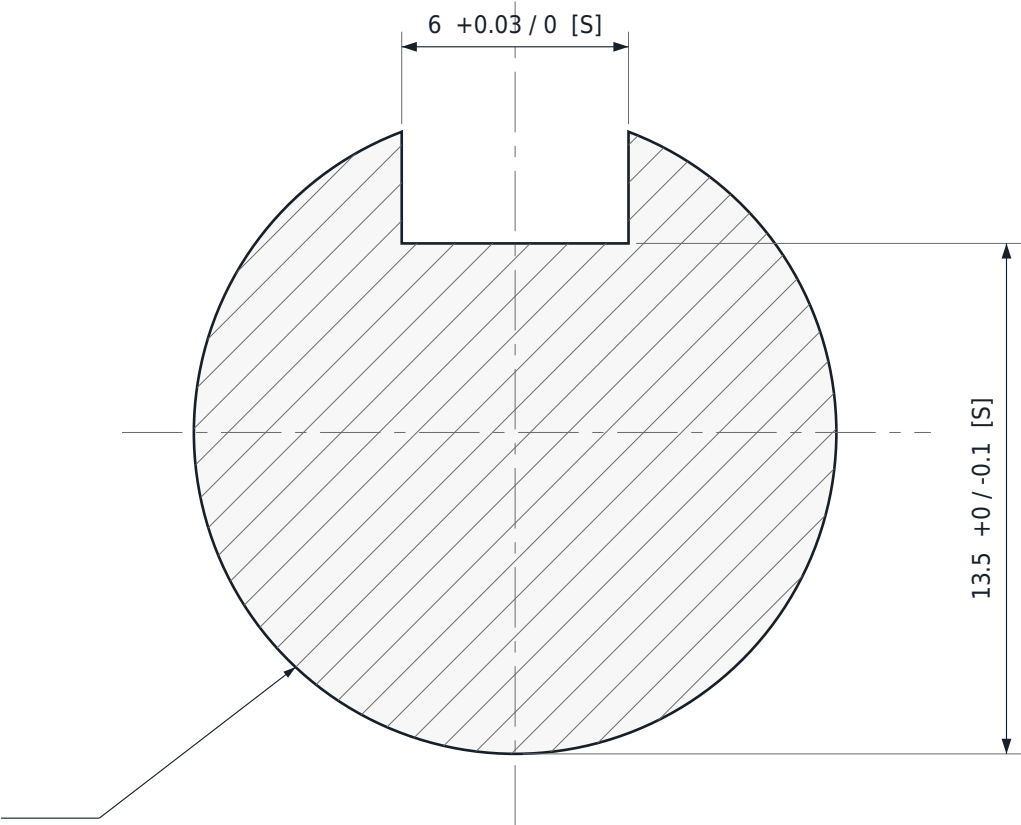
03 OUTPUT SHAFT / KEYWAY / END-FEATURE DETAILS

SC-MTR-1016Z-REF

REV A / 2026-10-01

A3 LANDSCAPE / ENGLISH

DETAIL K / SOURCE SHAFT CROSS-SECTION [S]

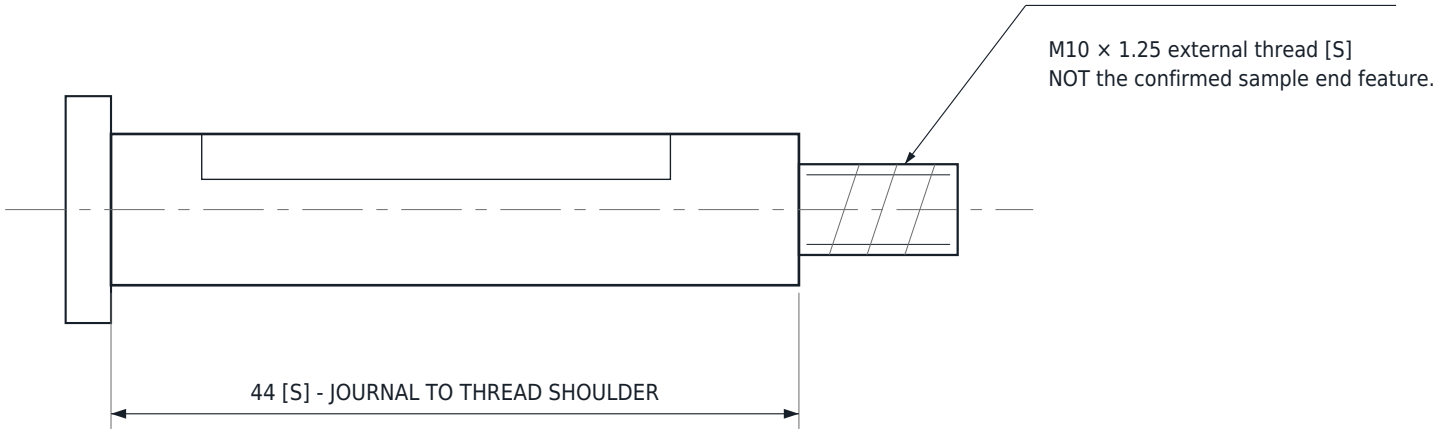


Ø17 -0.006 / -0.024 [S]
Nominal section scale: 5:1 on A3.

SOURCE FEATURE	NOMINAL / DEVIATIONS	DERIVED SOURCE LIMITS
Shaft diameter	Ø17 -0.006/-0.024	16.976 to 16.994
Keyway width	6 +0.03/0	6.000 to 6.030
Bottom-to-floor height	13.5 +0/-0.1	13.400 to 13.500
Nominal keyway depth	17 - 13.5 = 3.5 [D]	REF; not a new tolerance

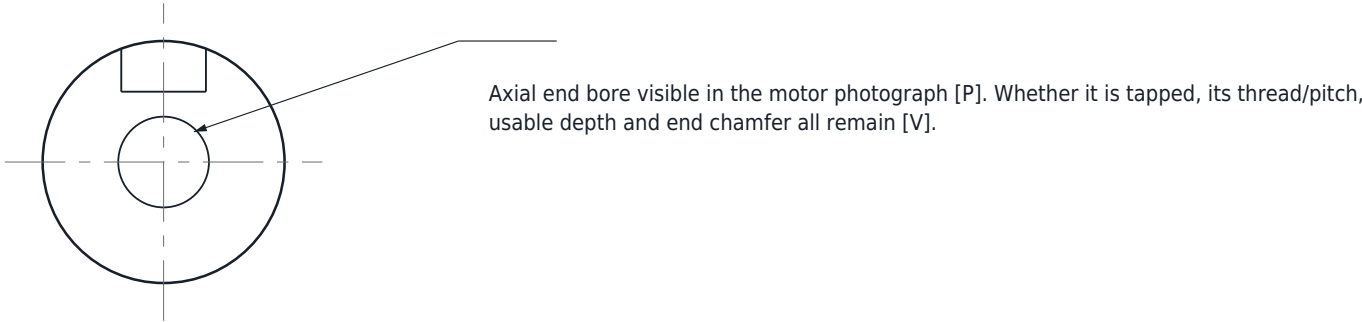
Only source tolerances are transcribed. The shaft diameter band is 0.018 mm wide; verify with an instrument and measurement process adequate for that band, not display resolution alone.

SOURCE SHAFT SIDE VIEW / VARIANT ONLY [S]



The source 44 mm ends at the start of the threaded tip; it is not the total shaft projection. Source thread length and total projection are not dimensioned.

ACTUAL END FEATURE / DO NOT SUBSTITUTE SOURCE THREAD



Keyway width, base height, length and angular position must also be confirmed. The 5:1 detail is a source section, not a certified fit specification for the KW1016Z sample.

[P] PHOTO / NAMEPLATE [S] SUPPLIED SOURCE VARIANT [D] DERIVED [V] VERIFY ACTUAL

Units: mm. No general tolerance is assigned. Do not scale NTS views or photographs.

REFERENCE / VERIFICATION REQUIRED - NOT RELEASED FOR MACHINING

No specimen caliper readings were supplied. [S] values are not measured KW1016Z dimensions.

Both purchased motors use this same part definition; installation rotation is an assembly task.

SHAFT AND KEYWAY

Single-hand motor reference / Smart-Cart Capstone

DRAWING SC-MTR-1016Z-REF	REVISION A	FORMAT A3	SHEET 3 / 5
VIEW SCALE: 5:1 / NTS	mm	EN	QTY 2

MY1016Z-FAMILY / 250 W DC GEAR MOTOR

Actual sample: KW1016Z-250W24V | ONE COMMON HAND / QUANTITY 2

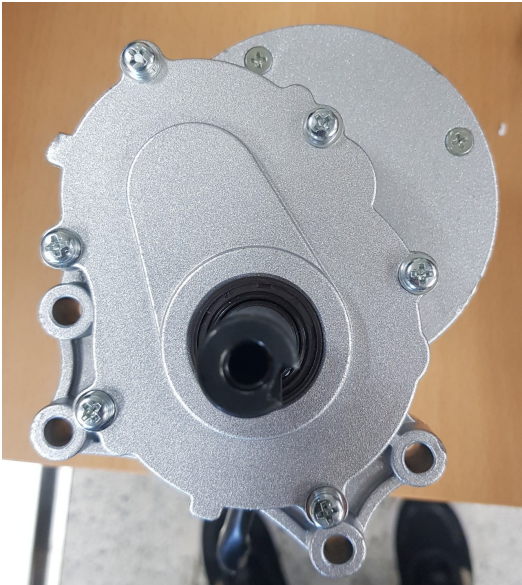
04 SOURCE RECONCILIATION / PHOTO EVIDENCE

SC-MTR-1016Z-REF

REV A / 2026-10-01

A3 LANDSCAPE / ENGLISH

ACTUAL PHOTOGRAPHS / GEOMETRY EVIDENCE [P]



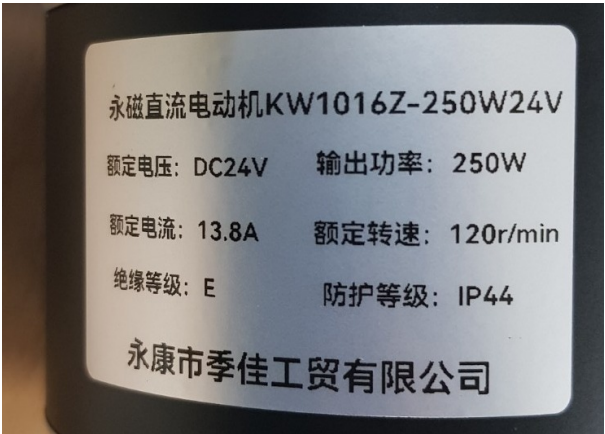
Output end / four mounting ears



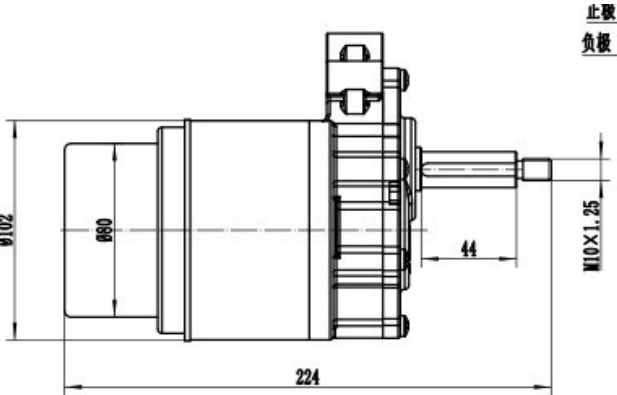
Side / unextended rear housing



Rear / no visible release lever



Nameplate: KW1016Z-250W24V; 24 V; 250 W; 13.8 A; rated 120 r/min; E; IP44.



Supplied source: ZD101AZ-WX-02, L (250W-24V). Rear attachment and external threaded tip are visible; the full 224 mm dimension belongs to that source configuration.

SOURCE-VARIANT APPLICABILITY

FEATURE	ACTUAL SAMPLE [P]	SOURCE ONLY [S]
Identity	KW1016Z-250W24V	ZD101AZ-WX-02; L
Voltage / output	24 V / 250 W	24 V / 250 W
Current	13.8 A, rated	13.4 A, rated
Speed	120 r/min, rated	110 rpm, no load
Motor body Ø	Measure actual [V]	102 [S]
Rear attachment Ø	Not applicable as drawn	80 [S]; rear attachment
Overall axial length	Measure actual [V]	224 [S]; full source unit
End feature	Axial bore visible [P]	M10 × 1.25, external [S]
Journal length	Measure actual [V]	44 [S]; excludes tip
Rear brake / lever	Not visible in photos	Attachment / lever shown
Lead specification	Size / length [V]	2 mm²; 300 exposed [S]
Mounting holes	Verify on both [V]	4 × Ø6.5 [S]

WHAT THIS REVISION DOES NOT CLAIM

The source drawing is not an exact drawing of the purchased sample. Overall 224, rear Ø80, external M10 thread, 110 rpm no-load speed and 13.4 A must not overwrite the physical sample data. No unmeasured envelope or torque is approved here.

EVIDENCE REGISTER

[P1] User motor photographs: 20261001_113605 / 113630 / 113653 / 113711 / 113731.
[S1] User-supplied partial source drawing: image(1).png. English callouts in this PDF are transcriptions.
[R1] Project V8 3D/CAD specification (2026-09-21) / component-register.csv: no actual caliper dataset.

[P] PHOTO / NAMEPLATE [S] SUPPLIED SOURCE VARIANT [D] DERIVED [V] VERIFY ACTUAL

Units: mm. No general tolerance is assigned. Do not scale NTS views or photographs.

REFERENCE / VERIFICATION REQUIRED - NOT RELEASED FOR MACHINING

No specimen caliper readings were supplied. [S] values are not measured KW1016Z dimensions.

Both purchased motors use this same part definition; installation rotation is an assembly task.

EVIDENCE AND APPLICABILITY

Single-hand motor reference / Smart-Cart Capstone

DRAWING SC-MTR-1016Z-REF	REVISION A	FORMAT A3	SHEET 4 / 5
VIEW SCALE: NTS	mm	EN	QTY 2

MY1016Z-FAMILY / 250 W DC GEAR MOTOR

Actual sample: KW1016Z-250W24V | ONE COMMON HAND / QUANTITY 2

05 PHYSICAL VERIFICATION / TWO IDENTICAL SPECIMENS

SC-MTR-1016Z-REF

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A3 LANDSCAPE / ENGLISH

Purpose: complete one verified part record for two same-handed specimens. Use the coordinate system on Sheet 02. Record actual units and instrument identity; a blank cell is not an accepted dimension.

SPECIMEN A ID: _____ SPECIMEN B ID: _____ INSTRUMENT / SERIAL / ACCURACY: _____

ID	ACTUAL FEATURE TO RECORD	REFERENCE, NOT ACCEPTANCE	SPECIMEN A	SPECIMEN B	METHOD / INTERFACE NOTE
D01	Motor cylinder outside diameter	102 [S]			Outside measurement
L01	Complete axial length, back to shaft tip	Actual [V]; not 224			Exclude cable; include end features
L02	Mounting contact plane A to shaft tip	Actual [V]			Pad plane must be identified
L03	Motor can length / rear cap length	Actual [V]			Record lengths separately
L04	Gearcase axial thickness / front boss projection	Actual [V]			No source axial dimensions
E01	Maximum transverse width × height	Actual [V]			Include ears and cover screw heads
E02	Can-axis offset from output axis (X, Y)	Actual [V]			Record signed components
D02	Output shaft diameter	17 [S]; limits on 03			Measure at several axial / angular sites
K01	Keyway width / bottom-to-floor height	6 / 13.5 [S]			Check source tolerances separately
K02	Keyway axial length / angle to X-axis	Actual [V]			Angle is not defined by source section
T01	End bore: plain / tapped; diameter or thread	Actual [V]			Do not assume source M10 × 1.25
T02	Bore usable depth / thread engagement	Actual [V]			Use depth and thread gauges as needed
H01	Mounting bore diameters H1 / H2 / H3 / H4	4 × 6.5 [S]			Four bores; not the cover fasteners
H02	H1 coordinate (X, Y)	(-24.5, +42.5) [S]			Use common datums and clocking
H03	H2 coordinate (X, Y)	(+2.0, +55.0) [S]			Do not round X offset to zero
H04	H3 coordinate (X, Y)	(+54.7, +2.0) [S]			Do not round Y offset to zero
H05	H4 coordinate (X, Y)	(+41.8, -24.5) [S]			Y from shaft horizontal axis
H06	Six hole-to-hole center distances	Sheet 02 [D]			Independent closure check
M01	Pad / ear thickness; face coplanarity	Actual [V]			Confirm all four contact surfaces
M02	Mounting-bolt access / cover-head projection	Actual [V]			Include washer and tool keepout
C01	Cable exit; free length; connector and bend space	Actual [V]			Source 2 mm² / 300 is not confirmed
W01	Assembly mass	Actual [V]			Weigh with defined cable inclusion

Measurement consistency: for two equal bores of diameter d, center distance = outer span - d = inner gap + d. For unequal bores, use half the sum of their diameters. Check all six distances; record values rather than forcing them to match the source.

Release criteria: verify both specimens, resolve source/part mismatches, update all [V] entries, check hub retention and bracket clearances, then issue a new revision. This PDF does not authorize changes to the gearbox or safety systems.

RECORDED BY: _____ DATE: _____ REVIEWED BY: _____ NEXT REVISION / STATUS: _____

[P] PHOTO / NAMEPLATE [S] SUPPLIED SOURCE VARIANT [D] DERIVED [V] VERIFY ACTUAL Units: mm. No general tolerance is assigned. Do not scale NTS views or photographs. REFERENCE / VERIFICATION REQUIRED - NOT RELEASED FOR MACHINING No specimen caliper readings were supplied. [S] values are not measured KW1016Z dimensions. Both purchased motors use this same part definition; installation rotation is an assembly task.	MEASUREMENT RECORD Single-hand motor reference / Smart-Cart Capstone			
	DRAWING SC-MTR-1016Z-REF	REVISION A	FORMAT A3	SHEET 5 / 5
	VIEW SCALE: NTS	mm	EN	QTY 2